GEOTECHNICAL INVESTIGATION REPORT

Sage Creek Main Canal — Reach 7 Rehabilitation
Station 142+00 to 166+00

Magic Valley, Twin Falls County, Idaho

Prepared for:
Sage Creek Irrigation District

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1.0 INTRODUCTION

High Desert Engineering, LLC (HDE) performed a geotechnical investigation for the planned rehabilitation of Reach 7 (Sta. 142+00 to 166+00) of the Sage Creek Main Canal. This investigation was recommended by the 2023 Canal Condition Assessment (HDE, September 2023) to characterize the canal subgrade, bank materials, and the active right-bank slope failure prior to rehabilitation design. The investigation addresses three primary concerns: (1) expansive clay subgrade causing lining panel heaving, (2) right bank instability, and (3) seepage pathways through the soil profile.

2.0 FIELD EXPLORATION

2.1 Drilling Program

Four exploratory borings were drilled between January 8 and January 15, 2025, during the non-irrigation season with the canal dewatered. Drilling was performed by Snake River Drilling, LLC using a truck-mounted CME-75 drill rig. All borings were drilled from the dewatered canal prism or adjacent bank using 4.25-inch hollow-stem augers with standard penetration test (SPT) sampling and Shelby tube sampling for undisturbed specimens.

Table 2-1: Boring Program Summary

Boring	Location	Elev. (ft)	Depth (ft)	GW (ft bgs)	Purpose
BH-SC-01	Canal invert, Sta. 153+00 (slide area)	4,285.0	30.0	8.5	Slide foundation, expansive clay
BH-SC-02	Right bank crest, Sta. 153+00 (slide head scarp)	4,297.0	40.0	14.0	Slide mass, failure surface
BH-SC-03	Canal invert, Sta. 149+00 (heave area)	4,286.5	25.0	9.0	Expansive subgrade under lining
BH-SC-04	Left bank crest, Sta. 149+00 (stable)	4,292.5	20.0	12.5	Control boring, baseline

2.2 Subsurface Conditions

The subsurface conditions encountered in the four borings are consistent with the regional geology described in the 2023 assessment. The generalized profile at the canal is:

- **Unit 1 — Loess (ML to CL):** Tan to brown, wind-deposited silt to silty clay, 5 to 12 feet thick. Dry density 82 to 90 pcf (loose for a silt), natural moisture content 10 to 14 percent. Very stiff when dry, but structurally metastable — collapses when wetted under load. SPT N-values 8 to 18 (medium to stiff).
- **Unit 2 — Lacustrine Clay (CH):** Green-gray to olive, high-plasticity clay, 6 to 14 feet thick. LL 55 to 68, PI 30 to 42. Stiff to very stiff (N = 15 to 28). Moisture content 28 to 35 percent. Slickensided joint surfaces, calcareous nodules, and oxidation staining along fractures. This is the primary expansive unit causing lining heave and the failure surface for the right bank slide.
- **Unit 3 — Alluvial Sand (SP to SP-SM):** Discontinuous lenses of poorly graded sand, 0 to 4 feet thick, encountered between the loess and clay units in BH-SC-01 and BH-SC-03. Loose to medium dense (N = 6

to 12). Saturated where present. Acts as a preferential seepage conduit.

- **Unit 4 — Basalt Bedrock:** Encountered at 22 to 28 feet in all borings. Upper 2 to 3 feet vesicular and fractured, then grading to massive. Auger refusal or very slow penetration in all borings. Not cored for this investigation.

2.3 In-Situ Permeability (Slug Tests)

Rising-head slug tests were performed in two borings (BH-SC-01 and BH-SC-03) using temporary well screens installed in the alluvial sand lenses. Results were analyzed using the Bouwer-Rice method.

Table 2-2: Slug Test Permeability Results

Boring	Screen Interval (ft)	Formation	k (cm/s)	k (ft/day)	Classification
BH-SC-01	10-14	Alluvial sand (SP)	$3.5 \times 10^{\blacksquare 3}$	9.9	Pervious
BH-SC-01	16-20	Lacustrine clay (CH)	$8.2 \times 10^{\blacksquare\blacksquare}$	$2.3 \times 10^{\blacksquare\blacksquare}$	Practically impervious
BH-SC-03	8-12	Alluvial sand / loess	$1.8 \times 10^{\blacksquare 3}$	5.1	Pervious

The permeability contrast between the alluvial sand (k ~ $10^{\blacksquare 3}$ cm/s) and the lacustrine clay (k ~ $10^{\blacksquare\blacksquare}$ cm/s) is approximately five orders of magnitude. This confirms that the sand lenses serve as the primary seepage pathways, while the clay acts as an aquitard. Canal seepage enters through the damaged lining, infiltrates vertically through the loess (moderate permeability, k ~ $10^{\blacksquare\blacksquare}$ to $10^{\blacksquare\blacksquare}$ cm/s), and flows laterally through the sand lenses, saturating the bank materials and driving the slope failure.

3.0 LABORATORY TESTING

3.1 Index Properties — Loess

Table 3-1: Loess — Index Properties and Gradation

Sample	Boring	Depth (ft)	LL	PL	PI	USCS	MC (%)	γd (pcf)	Gravel (%)	Sand (%)	Fines (%)
SC01-SS2	BH-SC-01	3-5	28	22	6	ML	11.8	86.5	0	12	88
SC01-SS3	BH-SC-01	5-7	31	23	8	CL	12.5	84.2	0	10	90
SC02-SS1	BH-SC-02	2-4	26	21	5	ML	10.2	89.5	0	14	86
SC02-SS3	BH-SC-02	8-10	30	22	8	CL	13.0	83.8	0	11	89
SC03-SS1	BH-SC-03	2-4	27	22	5	ML	11.0	87.8	0	13	87
SC04-SS1	BH-SC-04	2-4	29	23	6	ML	10.5	88.2	0	12	88
SC04-SS2	BH-SC-04	6-8	32	23	9	CL	13.8	82.5	0	8	92

3.2 Index Properties — Lacustrine Clay

Table 3-2: Lacustrine Clay — Index Properties

Sample	Boring	Depth (ft)	LL	PL	PI	USCS	MC (%)	γd (pcf)	Clay (%)	Activity
SC01-SS5	BH-SC-01	12-14	58	24	34	CH	31.2	98.5	48	0.71
SC01-SS7	BH-SC-01	18-20	62	25	37	CH	33.5	96.2	52	0.71
SC02-SS5	BH-SC-02	14-16	55	23	32	CH	28.8	100.5	45	0.71
SC02-SS7	BH-SC-02	20-22	65	24	41	CH	35.0	94.8	55	0.75
SC02-SS9	BH-SC-02	28-30	68	26	42	CH	34.2	95.5	58	0.72
SC03-SS4	BH-SC-03	10-12	60	25	35	CH	30.5	97.8	50	0.70
SC03-SS6	BH-SC-03	16-18	64	24	40	CH	33.8	95.0	54	0.74
SC04-SS4	BH-SC-04	10-12	57	23	34	CH	29.5	99.2	47	0.72

The lacustrine clay is classified as CH (fat clay) in all borings, with liquid limits ranging from 55 to 68, plasticity indices from 32 to 42, and clay fractions of 45 to 58 percent. The activity ratios of 0.70 to 0.75 are consistent with montmorillonite-dominant mineralogy, which is the primary swelling clay mineral. These properties place the material in the "high" to "very high" swell potential category per the USBR classification system.

3.3 Swell Pressure and Free Swell

Table 3-3: Swell Pressure and Free Swell (ASTM D4546 Method A)

Sample	Boring	Depth (ft)	Initial MC (%)	Final MC (%)	Swell Pressure (psf)	Free Swell (%)	Swell Potential
SC01-ST1	BH-SC-01	13-15	30.5	38.2	4,200	5.8	High
SC01-ST2	BH-SC-01	18-20	33.0	40.5	5,800	7.2	Very High
SC02-ST1	BH-SC-02	15-17	28.5	36.0	3,600	4.5	High
SC02-ST2	BH-SC-02	22-24	34.5	42.8	6,500	8.1	Very High
SC03-ST1	BH-SC-03	11-13	29.8	37.5	4,800	6.2	High
SC03-ST2	BH-SC-03	16-18	33.2	41.0	5,500	7.0	Very High
SC04-ST1	BH-SC-04	10-12	29.0	36.5	3,800	4.8	High

Swell pressures range from 3,600 to 6,500 psf (25 to 45 psi), increasing with depth as the clay becomes more plastic and drier (farther below the zone of seasonal moisture fluctuation). Free swell values of 4.5 to 8.1 percent confirm the high expansion potential. For a 4-inch-thick concrete lining panel sitting on 8 feet of this clay, the potential uplift force is approximately 2,000 to 3,200 lbs per square foot of panel — far exceeding the self-weight of the panel (approximately 50 psf). This explains the pervasive panel heaving observed in the condition assessment.

3.4 Collapse Potential — Loess

Table 3-4: Collapse Potential Test Results (ASTM D5333)

Sample	Boring	Depth (ft)	Initial e	Initial MC (%)	Collapse Potential (%)	Severity
SC01-CP1	BH-SC-01	3-5	0.82	11.8	8.5	Moderately severe
SC02-CP1	BH-SC-02	4-6	0.78	10.5	6.8	Moderate
SC02-CP2	BH-SC-02	8-10	0.85	13.0	10.2	Severe
SC03-CP1	BH-SC-03	3-5	0.80	11.0	7.5	Moderately severe
SC04-CP1	BH-SC-04	3-5	0.76	10.5	5.5	Moderate

Collapse potential values range from 5.5 to 10.2 percent, classified as "moderate" to "severe" per the Jennings and Knight (1975) classification. The highest collapse potential (10.2 percent, BH-SC-02 at 8-10 ft) is in the loess on the right bank where the slide head scarp is located. When canal seepage saturates this loess, the soil structure collapses under the self-weight of the overlying bank, creating the tension cracks and vertical displacement observed at the head scarp. This is a key mechanism in the slope failure.

3.5 Direct Shear Strength

Table 3-5: Direct Shear Test Results (ASTM D3080)

Sample	Boring	Depth (ft)	Material	c' (psf)	phi' (deg)	c'r (psf)	phi'r (deg)	Notes
SC01-DS1	BH-SC-01	4-6	Loess (ML)	180	28	0	26	Dry strength; collapses when wet
SC01-DS2	BH-SC-01	14-16	Lacust. clay (CH)	250	18	50	12	Slickensided surfaces
SC02-DS1	BH-SC-02	6-8	Loess (CL)	150	26	0	24	Pre-wetted specimen
SC02-DS2	BH-SC-02	16-18	Lacust. clay (CH)	280	19	60	13	Peak on intact, residual on slicks
SC02-DS3	BH-SC-02	24-26	Lacust. clay (CH)	320	20	80	14	Deepest, highest PI
SC04-DS1	BH-SC-04	4-6	Loess (ML)	200	29	0	27	Control boring
SC04-DS2	BH-SC-04	10-12	Lacust. clay (CH)	240	18	40	12	Similar to SC01

The lacustrine clay exhibits a significant strength reduction from peak to residual conditions: phi' drops from 18-20 degrees (peak) to 12-14 degrees (residual), and c' drops from 240-320 psf to 40-80 psf. The residual strength is controlled by the slickensided surfaces (pre-existing shear surfaces from tectonic and desiccation history). For the right bank slide, the failure surface passes through the clay along these pre-existing slickensides, and residual strength parameters should be used for back-analysis and remedial design.

The loess exhibits apparent cohesion of 150-200 psf when dry, which drops to zero upon saturation. The phi' angle (26-29 degrees) is relatively insensitive to moisture. The loss of apparent cohesion upon wetting is the trigger mechanism for the bank failure — when canal seepage saturates the loess, the apparent cohesion is lost and the only resistance to sliding is the frictional component.

3.6 Permeability — Laboratory

Table 3-6: Laboratory Permeability (ASTM D5084)

Sample	Boring	Depth (ft)	Material	Test Method	k (cm/s)	Notes
SC01-FH1	BH-SC-01	4-6	Loess (ML)	Falling head	4.5 × 10■■	Intact, unsaturated
SC01-FH2	BH-SC-01	14-16	Lacust. clay	Falling head	6.8 × 10■■	Very low, aquitard
SC02-FH1	BH-SC-02	6-8	Loess (CL)	Falling head	2.2 × 10■■	Slightly finer, lower k
SC03-FH1	BH-SC-03	4-6	Loess (ML)	Falling head	5.1 × 10■■	Similar to SC01
SC03-FH2	BH-SC-03	12-14	Lacust. clay	Falling head	7.5 × 10■■	Consistent with SC01

3.7 Moisture-Density Relationships

Table 3-7: Moisture-Density Relationships (ASTM D698)

Sample	Boring	Material	Test Method	OMC (%)	MDD (pcf)	Notes
SC01-PD1	BH-SC-01	Loess (ML)	Std. Proctor	16.5	106.2	Typical Magic Valley loess
SC01-PD2	BH-SC-01	Lacust. clay (CH)	Std. Proctor	24.0	100.5	High OMC due to clay content
SC02-PD1	BH-SC-02	Loess (CL)	Std. Proctor	17.2	104.8	Slightly finer loess
SC03-PD1	BH-SC-03	Lacust. clay (CH)	Std. Proctor	25.5	98.8	Highest PI specimen

3.8 Concrete Core Testing

Four concrete cores were extracted from representative lining panels at different condition levels for compressive strength and petrographic evaluation.

Table 3-8: Concrete Core Compressive Strength and Condition

Core ID	Station	Panel Condition	Thickness (in)	f'c (psi)	ASR	Freeze- Thaw
CC-1	145+20	Good	4.1	3,850	Minor	Minor scaling
CC-2	150+50	Fair	3.8	2,920	Moderate	Moderate popouts
CC-3	153+80	Poor	3.2	1,680	Severe	Severe delamination
CC-4	155+00	Failed	2.5	980	Severe	Disintegrated top 1 in

The concrete strength ranges from 980 psi (failed panels) to 3,850 psi (good panels). The original 28-day design strength was 3,000 psi. The good panels have gained strength over 61 years, but the poor and failed panels have lost 44 to 67 percent of their strength due to the combined effects of ASR and freeze-thaw degradation. The failed panels are structurally unsuitable for continued service; the 980 psi strength is below the minimum threshold for canal lining (approximately 2,500 psi per USBR standards).

4.0 ENGINEERING ANALYSIS

4.1 Right Bank Slope Stability

Slope stability analyses were performed for the right bank cross-section at Sta. 153+00 (the maximum bank height section through the slide) using Spencer's method. Two conditions were analyzed: (1) current condition with saturated loess and residual strength on the clay failure surface, and (2) remediated condition with bank re-grading and drainage.

Table 4-1: Right Bank Slope Stability Results

Condition	Phreatic Surface	Clay Strength	FS	USBR Minimum	Status
Existing (dry season)	Low (canal empty)	Peak	1.45	1.3	Adequate
Existing (irrigation)	High (canal full)	Peak	1.15	1.3	INADEQUATE
Existing (irrigation)	High (canal full)	Residual	0.92	1.3	FAILED (active slide)
Remediated: regrade to 3H:1V	Drainage blanket	Residual	1.35	1.3	Adequate
Remediated: regrade + geogrid	Drainage blanket	Residual	1.52	1.3	Adequate

The analysis confirms that the right bank is unstable during the irrigation season when the bank materials are saturated (FS = 0.92 using residual strength, consistent with the active slide). The failure surface passes through the loess (which loses its apparent cohesion when saturated) and along the slickensided clay surface at the loess/clay contact. Remediation requires: (1) re-grading the bank to 3H:1V (from the existing 2H:1V), (2) installing a drainage blanket behind the new lining to prevent saturation of the loess, and (3) optionally, reinforcing the slope with geogrid layers for additional margin.

4.2 Subgrade Swell Analysis

The swell potential of the lacustrine clay subgrade was evaluated for its impact on canal lining design. With swell pressures of 3,600 to 6,500 psf, the clay can exert uplift forces that exceed the weight of any practical unreinforced concrete lining. The following subgrade treatment alternatives were evaluated:

- **Option 1 — Overexcavation and replacement (2 ft):** Remove 2 feet of clay below the lining and replace with compacted non-expansive granular fill (pit-run gravel, GP-GM). This separates the lining from the active swell zone and provides a uniform bearing surface. Estimated to reduce swell-induced stresses on the lining by 80 to 90 percent.
- **Option 2 — Lime treatment (8 inches):** Scarify the upper 8 inches of subgrade, mix with 5 to 7 percent hydrated lime, and compact. Lime stabilization permanently modifies the clay mineralogy by cation exchange and pozzolanic reaction, reducing the PI from 30-42 to typically 8-15 and eliminating swell potential. Requires a 7-day curing period.
- **Option 3 — Geomembrane barrier:** Install an HDPE or PVC geomembrane between the subgrade and lining. The geomembrane prevents moisture exchange between the canal water and the clay subgrade, limiting the wetting cycles that drive swell. Less effective than Options 1 or 2 at reducing swell, but provides excellent seepage control.

Table 4-2: Subgrade Treatment Alternatives Comparison

Treatment	Swell Reduction	Seepage Reduction	Cost ($/SF)	Durability	Rating
2 ft overexcavation + granular backfill	80-90%	Moderate (granular is permeable)	$6-8	40+ years	Preferred for swell control
8 in lime treatment	90-95%	Low (no barrier)	$3-5	30+ years	Cost-effective, proven locally
Geomembrane (40 mil HDPE)	30-50%	95%+	$4-6	25-30 years	Best seepage control
Combination: lime + geomembrane	90-95%	95%+	$7-10	30+ years	RECOMMENDED (comprehensive)

Recommendation: The combination of lime treatment (to control swell) with a geomembrane underliner (to control seepage) is recommended as the most effective subgrade treatment for the Reach 7 rehabilitation. This addresses both primary subgrade issues: expansive clay swell and seepage loss through the soil profile. The lime treatment is particularly well-suited to this project because extensive experience with lime-treated expansive clays exists in the Magic Valley region (ACHD and ITD have established specifications for local conditions).

4.3 Seepage Analysis

A steady-state seepage analysis was performed using SEEP/W (GeoStudio) to model flow through the canal cross-section under current (damaged lining) and rehabilitated conditions. Key findings:

- **Current condition:** Modeled seepage of 15.8 cfs per 1,000 ft (using equivalent permeability of the damaged lining = 10■■ cm/s), consistent with measured losses of 7.7 cfs per 1,000 ft (the model overestimates because it assumes uniform damage; actual damage is concentrated in 60% of the reach).
- **New PCC lining only (no geomembrane):** Modeled seepage of 0.8 cfs per 1,000 ft, a 95% reduction from current conditions. Meets USBR target of less than 1.0 cfs per 1,000 ft.
- **Geomembrane + PCC composite:** Modeled seepage of 0.05 cfs per 1,000 ft, a 99.7% reduction. Essentially eliminates seepage losses. This option also eliminates the wetting of the bank loess that drives the slope instability, providing a secondary benefit for bank stability.

5.0 CONCLUSIONS AND RECOMMENDATIONS

5.1 Conclusions

Expansive Subgrade: The lacustrine clay (CH) beneath the canal lining has swell pressures of 3,600 to 6,500 psf, which are the primary cause of lining panel heaving and failure. Any new lining must address the expansive subgrade to avoid repeating the same failure mechanism within its design life.

Bank Instability: The right bank slide is caused by loss of apparent cohesion in the loess when saturated by canal seepage, combined with a low-strength slickensided clay failure surface. The slide is active (FS = 0.92 during irrigation). Bank re-grading to 3H:1V with a drainage blanket is required before relining.

Seepage: Canal seepage enters through damaged lining, infiltrates the loess (k ~ 10■■ cm/s), and flows laterally through sand lenses (k ~ 10■^{3} cm/s). The seepage not only wastes water (5,400 AF/yr, $270,000/yr) but also saturates bank materials and drives instability. A geomembrane underliner virtually eliminates this mechanism.

Collapsible Loess: The loess deposits on the right bank have collapse potentials of 5.5 to 10.2 percent (moderate to severe). Collapse settlement contributes to the bank failure mechanism. The drainage blanket in the remediated section will prevent wetting of the loess and eliminate collapse as a concern.

Concrete Lining: The existing 1962 lining has exhausted its useful life. Compressive strengths in damaged panels (980 to 1,680 psi) are below the USBR minimum of 2,500 psi. Full removal and replacement is required.

5.2 Recommended Rehabilitation Design

Based on the investigation results, HDE recommends the following rehabilitation design for Reach 7:

- **Right bank:** Re-grade from 2H:1V to 3H:1V, install 12-inch drainage blanket (crushed gravel) behind lining, with perforated pipe drain at bank toe discharging to the canal downstream.
- **Subgrade treatment:** Lime-treat upper 8 inches of exposed subgrade at 6 percent lime by dry weight. Compact to 95 percent of Standard Proctor MDD. Allow 7-day cure before geomembrane placement.
- **Geomembrane:** Install 40-mil HDPE geomembrane over entire canal section (invert and both banks) with 6-inch overlap at seams, thermally welded. Provide 6-inch granular cushion below and above geomembrane.
- **New lining:** 4-inch reinforced concrete lining (4,000 psi, 6x6-W2.9xW2.9 welded wire fabric) placed over the geomembrane/cushion system. Use non-reactive aggregate (tested per ASTM C1260) and Type II low-alkali cement. Transverse contraction joints at 12-foot spacing with PVC waterstop.
- **Freeze-thaw protection:** Specify air-entrainment (6 ± 1 percent) and a maximum water-cement ratio of 0.45 for freeze-thaw durability in the Magic Valley climate (ASTM C260).

5.3 Updated Cost Estimate

Table 5-1: Detailed Rehabilitation Cost Estimate

Item	Quantity	Unit Cost	Total	Notes
Demolition / removal	28,800 SF	$3.50/SF	$101,000	Existing lining and debris
Right bank re-grading	2,400 CY	$18/CY	$43,000	Cut to 3H:1V, haul and waste
Drainage blanket	4,300 SF	$8/SF	$34,000	12-in crushed gravel + perf pipe

Item	Quantity	Unit Cost	Total	Notes
Lime treatment	2,140 CY	$22/CY	$47,000	6% lime, 8-in depth, full subgrade
Geomembrane (40 mil HDPE)	32,500 SF	$3.50/SF	$114,000	Incl. welding and QA
Granular cushion	1,800 CY	$28/CY	$50,000	6-in above and below membrane
Reinforced concrete lining	28,800 SF	$14/SF	$403,000	4 in, 4,000 psi, WWF
Joints and waterstop	2,400 LF	$12/LF	$29,000	PVC waterstop at 12-ft spacing
Mobilization (8%)	LS		$66,000	
Contingency (15%)	LS		$133,000	

Total estimated project cost: $1,020,000. This is within the range of the preliminary estimate ($849,000 to $1,417,000) from the 2023 assessment, at approximately the midpoint. The annual savings from eliminated seepage losses ($270,000/yr) result in a simple payback period of approximately 3.8 years, making this a strongly positive investment for the District.

6.0 LIMITATIONS

This report has been prepared for the exclusive use of the Sage Creek Irrigation District for the specific project described. Subsurface conditions were characterized at four boring locations; conditions may vary between borings. If conditions different from those described are encountered during construction, HDE should be notified immediately. The cost estimates are based on 2025 unit prices and are subject to revision based on bidding conditions.